# CodeMigrate: Remaining Work

For Prabhas, Sandhani, Dhanush and Hemanth. The code, results, slides and guides are ready. This page lists what is left.

**Read first:** `docs/CodeMigrate_Team_Guide.pdf` (your part's pages have a simple explanation, a code walkthrough, numbers to remember and likely questions with answers). **Paper help:** `paper/WRITING_PLAN.md` gives one cue per sentence.

## 1. Your tasks

| Name | Learn + explain in review | Code to open if mam asks | Write in paper |
|---|---|---|---|
| **Prabhas** | Gemini reasoner + citation validator (main novelty); the novelty table (`docs/CodeMigrate_Novelty.pdf`) | `llm/reasoner.py`, `llm/prompts.py`, `validation/citation_validator.py` | Abstract, Introduction, Related Work, Method: validator |
| **Sandhani** | Code analysis + retrieval; benchmark, metrics and results | `analysis/ast_analyzer.py`, `retrieval/hybrid.py`, `evaluation/metrics.py` | Method: analysis + retrieval, Benchmark, Results |
| **Dhanush** | Knowledge base (310 Django changes from release notes) | `kb/schema.py`, `data/kb/django_changes.jsonl` | Knowledge Base, Conclusion |
| **Hemanth** | Web app, CLI, fixer; **runs the live demo** | `api/routes.py`, `cli.py`, `patching/fixer.py` | System implementation, Limitations |

All code paths are inside `src/codemigrate/`.

## 2. Work we do together

| Task | Owner | Done when |
|---|---|---|
| Learn your part and practise explaining it in 2 minutes | Everyone | You can explain your code files without reading |
| Write your paper section in your own words | Everyone | Section sent to Sandhani |
| Merge all sections in Overleaf and compile the PDF | Sandhani | One IEEE PDF with all sections |
| Add all names in the author line | Sandhani | Names + college in the PDF |
| Run the AI-content and plagiarism checks | Hemanth | Both reports under 10%, saved as PDF |
| Rewrite any part the checks flag | Owner of that section | Re-check passes |
| Practise the full talk + live demo once together | Everyone | 12 minutes, no stops |
| Record a backup demo video | Hemanth | Video on the laptop in case the live demo fails |

## 3. Order of work

| Step | What |
|---|---|
| Day 1 | Everyone reads their part in the team guide and opens their code files. |
| Day 2-3 | Everyone writes their paper section. Send it to Sandhani. |
| Day 4 | Sandhani merges in Overleaf. Hemanth runs both checks. Fix anything flagged. |
| Day 5 | Full practice of talk + demo together. Hemanth records the backup video. |

## 4. Paper rules (must follow)

- Write in **your own words**. No ChatGPT or other AI tools, and no paraphrasing tools: AI content must be under 10%.
- Do not copy from the team guide, slides, papers or websites: similarity must be under 10%.

- Use the numbers exactly as in `results/tables/` and the FACTS notes in `paper/main.tex`. Tables, figures and references are already in the template.
- Simple, short sentences are fine. It is a draft.

## 5. Run the demo (Hemanth)

- Build the web app once: `cd frontend; npm run build; cd ..`
- Start: `$env:WEB_DIR="frontend/dist"; .venv\Scripts\uvicorn codemigrate.api.main:app --port 8000`
- Open `http://localhost:8000`, register, create a project, versions 4.0 to 5.1, reasoner **offline**, tab Paste code, paste the demo code, click Scan.
- Show the findings, a source link, and the Updated code tab. Full steps and the code to paste: `docs/DEMO_SCRIPT.md`.

## 6. Bring to the review

- Laptop with the demo working + backup video
- Slides (`docs/CodeMigrate_Review.pptx`) and novelty PDF
- Paper PDF + AI-content report + plagiarism report